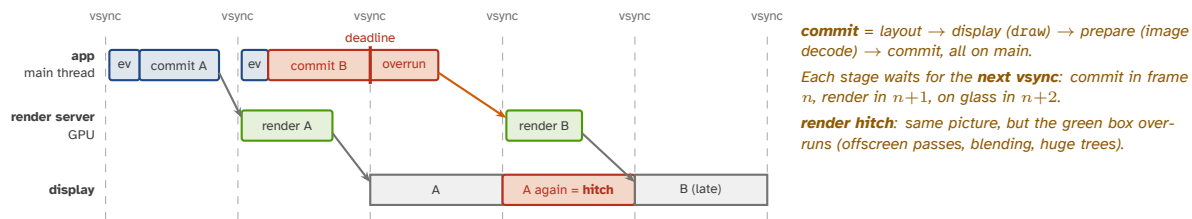


In one line: Your app never draws the screen: it edits a **layer tree** and **commits** it once per frame to the out-of-process **render server**, which renders it on the GPU for the next vsync. Every frame has a deadline — **16.67 ms at 60 Hz, 8.33 ms at 120 Hz (ProMotion)**. Miss it in *your* process → **commit hitch**; miss it in the render server → **render hitch**. Either way the old frame stays on screen.



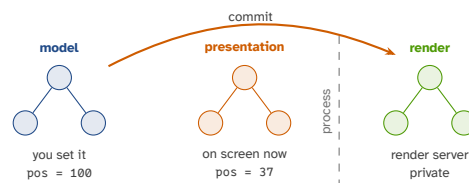
How it works

- Every **UIView** is backed by a **CALayer** (`view.layer`; the view is the layer's *delegate*; swap the type with `override class var layerClass`). View = events, responder chain, Auto Layout; layer = contents, geometry, animation.
- Three trees**: the **model** tree (what you set — target values), the **presentation** tree (`layer.presentation()`, in-flight values), the **render** tree (private, in the render server). Animations run in the render server — they keep going while your main thread is briefly busy.
- CATransaction** batches layer changes; the run loop opens an *implicit* one and commits it at the end of the turn. Explicit: `begin() / setDisableActions(true) / setAnimationDuration(_:) / setCompletionBlock / commit()`.
- Implicit animations**: setting an animatable property on a *standalone* layer animates (default action, default timing). A view's backing layer does *not* — **UIView** disables layer actions outside an animation block, so `view.layer.opacity = 0` snaps.
- Offscreen rendering** = GPU renders into an extra buffer, then composites it back (buffer switch per pass). Triggers: **shadow without shadowPath** (shape from alpha), **mask**, **cornerRadius + masksToBounds** over *content* (plain background + radius is fine), **group opacity** (alpha < 1 on a layer with sublayers), `shouldRasterize`.
- Rasterization** (`shouldRasterize = true`) caches the subtree as a bitmap: wins only for *static* complex content; if it changes each frame it re-renders offscreen every time. Set `rasterizationScale` to the screen scale (`traitCollection.displayScale`) or it blurs.
- Images decode lazily**, on *first display*, on the main thread (prepare phase). Memory = **px width × px height × 4**, not file size: $4000 \times 3000 \approx 48$ MB even in a 100 pt cell. Fix: **downsample** with `ImageIO`, or iOS 15 `prepareForDisplay()` (sync, call off main) / `prepareThumbnail(of:completionHandler:)` (background) — assign on main.

Example — downsample off the main thread

```
// maxPx = max(w, h) in points * displayScale -> PIXELS
func downsample(_ url: URL, maxPx: CGFloat) -> UIImage? {
    let noCache = [kCGImageSourceShouldCache: false] as CFDictionary
    guard let src = CGImageSourceCreateWithURL(url as CFURL, noCache)
    else { return nil }
    let opts = [kCGImageSourceCreateThumbnailFromImageAlways: true,
                kCGImageSourceShouldCacheImmediately: true,
                kCGImageSourceCreateThumbnailWithTransform: true,
                kCGImageSourceThumbnailMaxPixelSize: maxPx]
    as CFDictionary
    guard let cg = CGImageSourceCreateThumbnailAtIndex(src, 0, opts)
    else { return nil } // decoded NOW, at thumbnail size only
    return UIImage(cgImage: cg)
}
```

One layer, three trees



Offscreen triggers → fix

shadow, no path	<code>layer.shadowPath = UIBezierPath(...).cgPath</code>
radius + clip on an image	pre-round once when downsampling
mask layer	bake into the image / avoid in cells
alpha on a container	fade the leaves, or flatten
changing rasterized layer	turn <code>shouldRasterize</code> off

Traps

- ! "I have 16 ms" — not on ProMotion: 8.33 ms at 120 Hz.
- ! `cellForRow` fast but still hitching → the cost is in commit (decode, draw) or *render* (offscreen), not your code.
- ! `cornerRadius` alone is not the problem — radius *plus clipping content* is.
- ! `shouldRasterize` on animating content = worse, not better.
- ! Downsample to *points* → blurry: `MaxPixelSize` is pixels (× `displayScale`).
- ! Decode off main, but set `imageView.image` on main — **UIKit** is not thread-safe.
- ! Overriding `draw(_:)` allocates a backing store ($w \times h \times scale^2 \times 4$ B) and draws on the CPU; use layer properties.

Remember

"Set → commit → render → glass, one vsync each." Main thread late = commit hitch; GPU late = render hitch. Images: **pixels × 4**, decode once, small, off main.